



Comparative RNA-Seq Analysis of Gene Expression and RNA Modifications in Trunk and Limb Muscles of C57BL/6 Mice



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Abstract

Muscle specialization across anatomical regions is driven by distinct gene expression and RNA modification patterns [1,2]. To explore these regulatory mechanisms, the COSMOS Mortazavi Lab performed Oxford Nanopore direct RNA sequencing on trunk muscles (abdominal wall, pectoralis) from 10-week-old female mice and pulled data from the Mortazavi Lab of limb muscle (gastrocnemius) from 6-month-old male C57BL/6 mice. The study identified approximately 8994 genes with at least one shared inosine site, suggesting a role in core muscle functions [4]. Out of those, 763 genes shared at least three inosine sites across the trunk, limb, and total muscle groups. These findings highlight novel markers of muscle identity and demonstrate the utility of direct RNA sequencing in uncovering tissue-specific RNA regulation relevant to muscle development and disease [1].

Introduction

Skeletal muscle is a striated tissue responsible for voluntary movement, joint stability, and protein storage. It consists of long, multinucleated fibers containing sarcomeres, making it a valuable model for studying gene expression and RNA modifications. [5] Gene expression involves transcription and translation, processes essential for cellular function. [8] RNA modifications—such as m6A, m1A, m5C, and Ψ (pseudouridine)—regulate RNA stability, structure, and translation, forming a dynamic layer known as the epitranscriptome. [6] One key modification is the conversion of adenosine to inosine, which can alter protein coding, splicing, and RNA stability. [7] Investigating these changes in skeletal muscle reveals how post transcriptional regulation influences muscle function and identity.

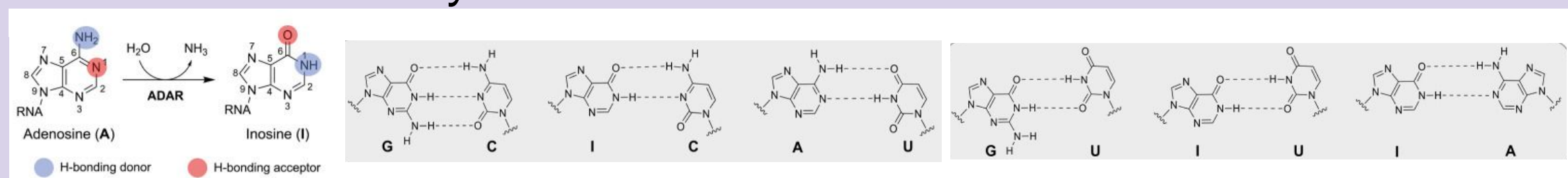


Figure 1: Inosine Modifications. The change of adenosine to inosine, altering RNA function [9].

Methodology

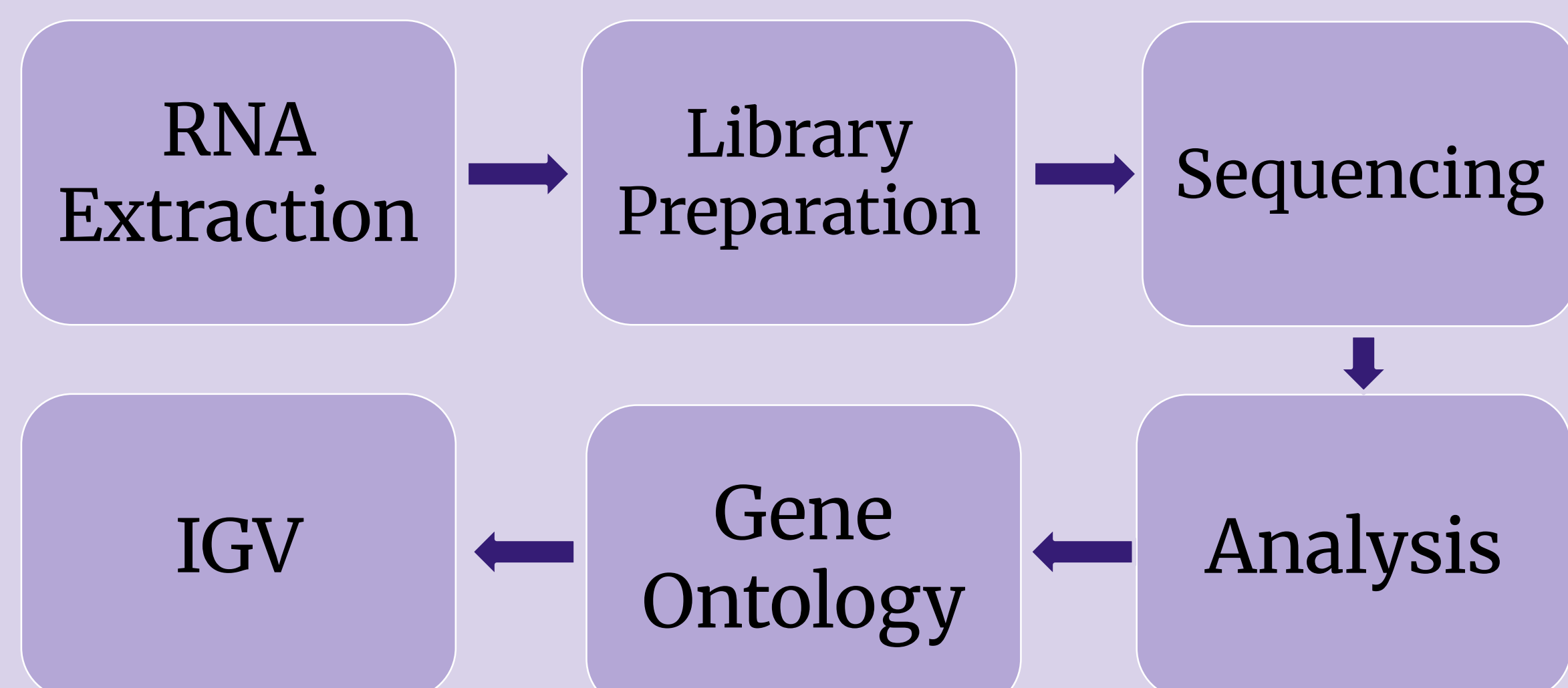


Figure 2: Overview of Methodologies. Steps of wet lab, processed data, and conclusion

Principal Component Analysis Shows Muscle Group Differences

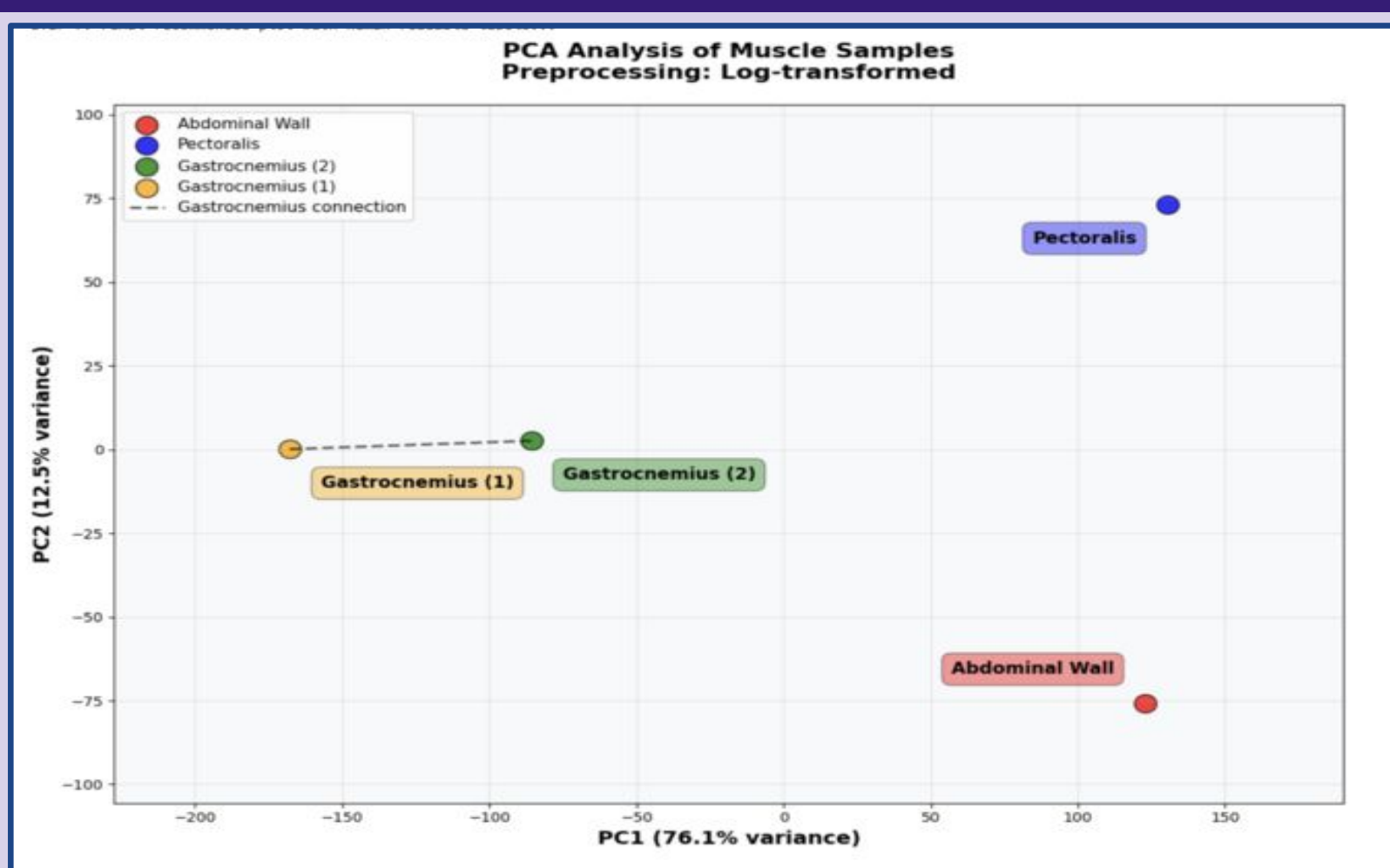


Figure 3: Principal Component Analysis of Muscle Samples. Plot showing gene expression profiles that separate trunk muscles (abdominal wall, pectoralis) from limb muscles (gastrocnemius 1 and gastrocnemius 2) along PC1 and PC2, highlighting distinct transcriptomic signatures for each muscle group

Differentially Expressed Gene Analysis of Trunk vs. Limb

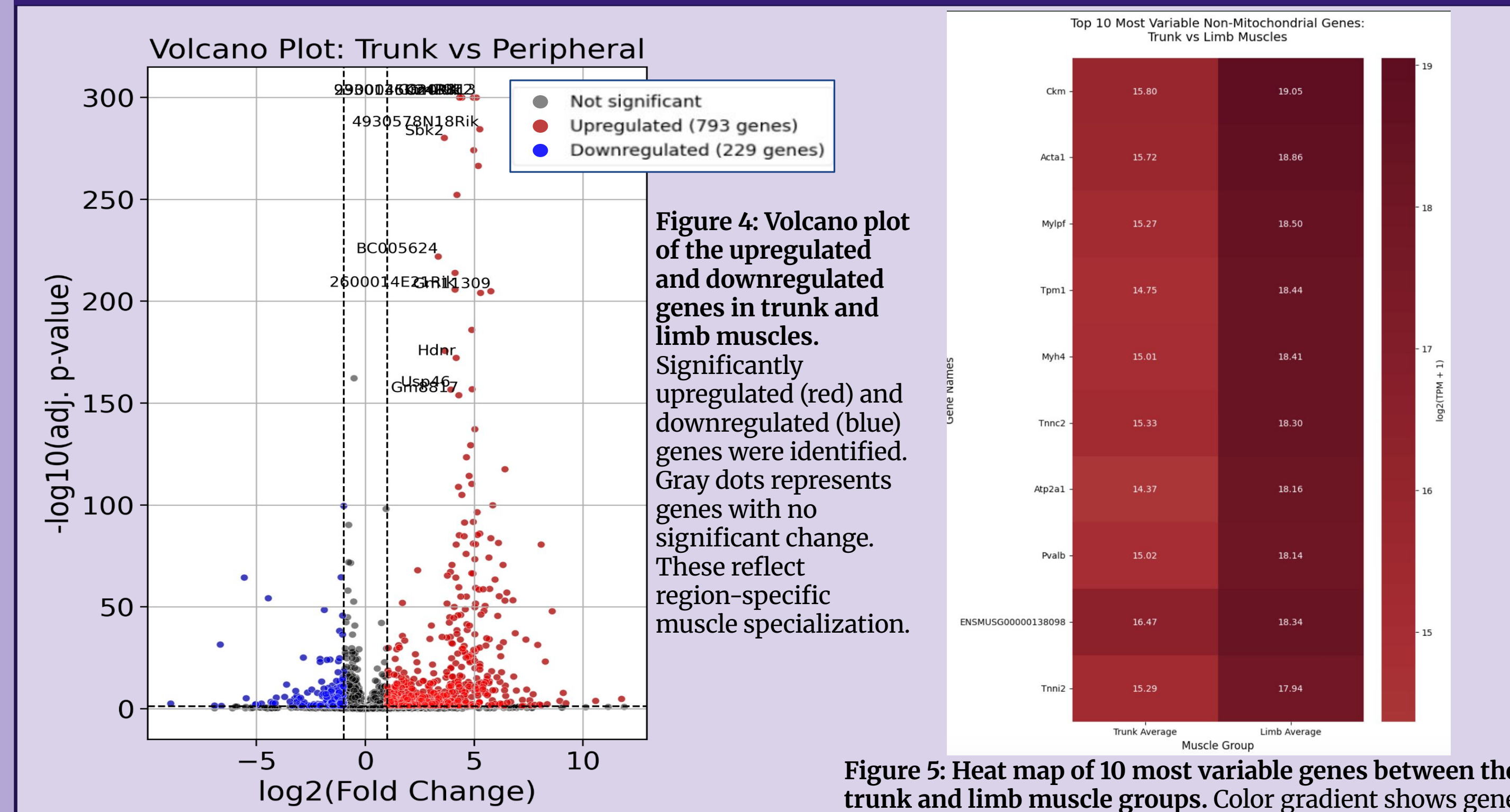


Figure 5: Heat map of 10 most variable genes between the trunk and limb muscle groups. Color gradient shows gene expression. Limb muscles have greater gene expression.

Gene Ontology Analysis of Differentially Expressed Genes: Differences in Thermogenesis And Channel Activity

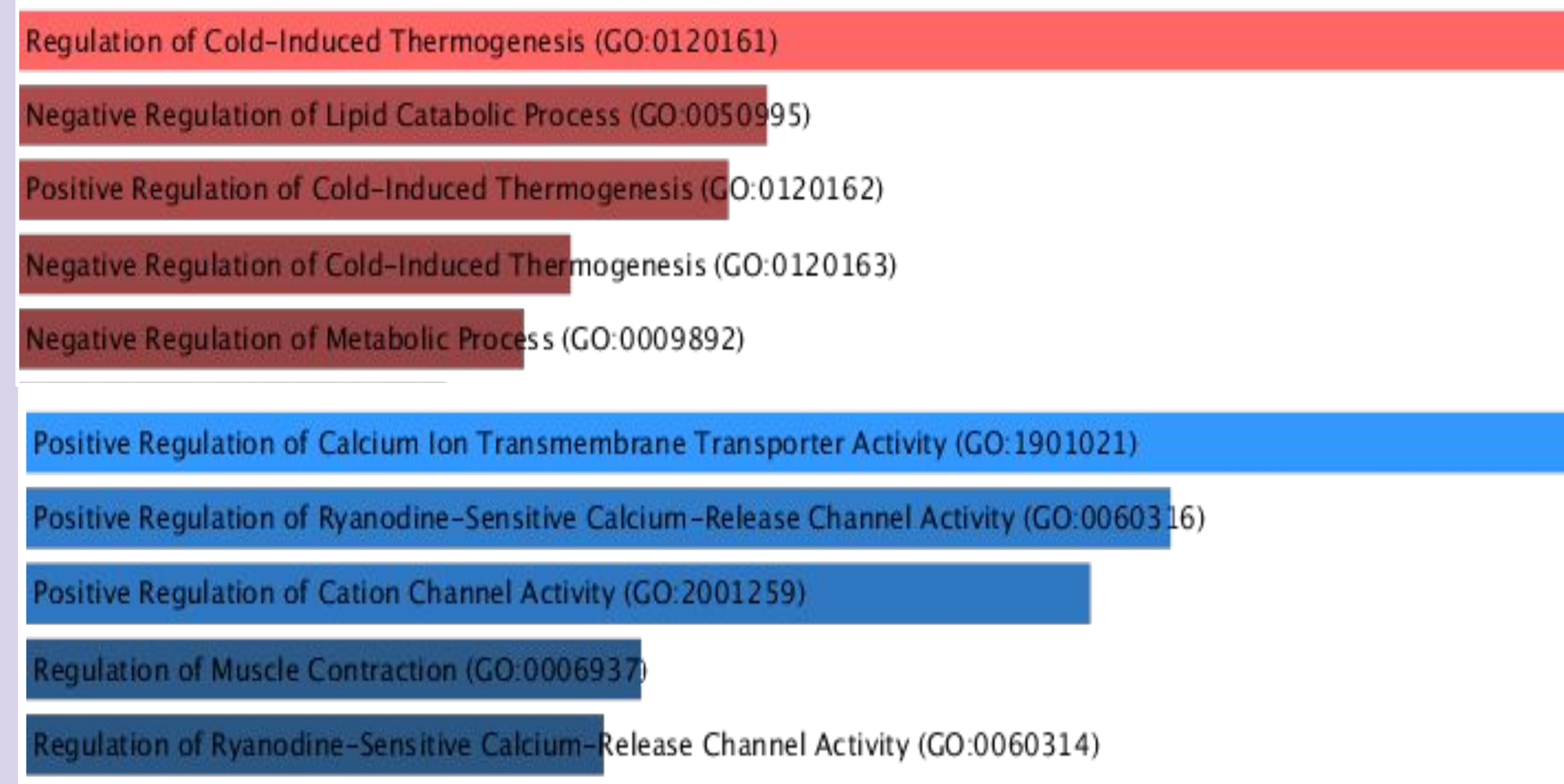


Figure 6: GO analysis of the non mitochondrial upregulated genes explaining the biological processes prevalent in the genes when comparing limb upregulated to trunk downregulated

Figure 7: GO analysis of the non mitochondrial downregulated genes explaining the biological processes prevalent in the genes when comparing trunk upregulated to limb downregulated

Analysis of Shared Inosine Sites Across Trunk And Limb Muscle Samples

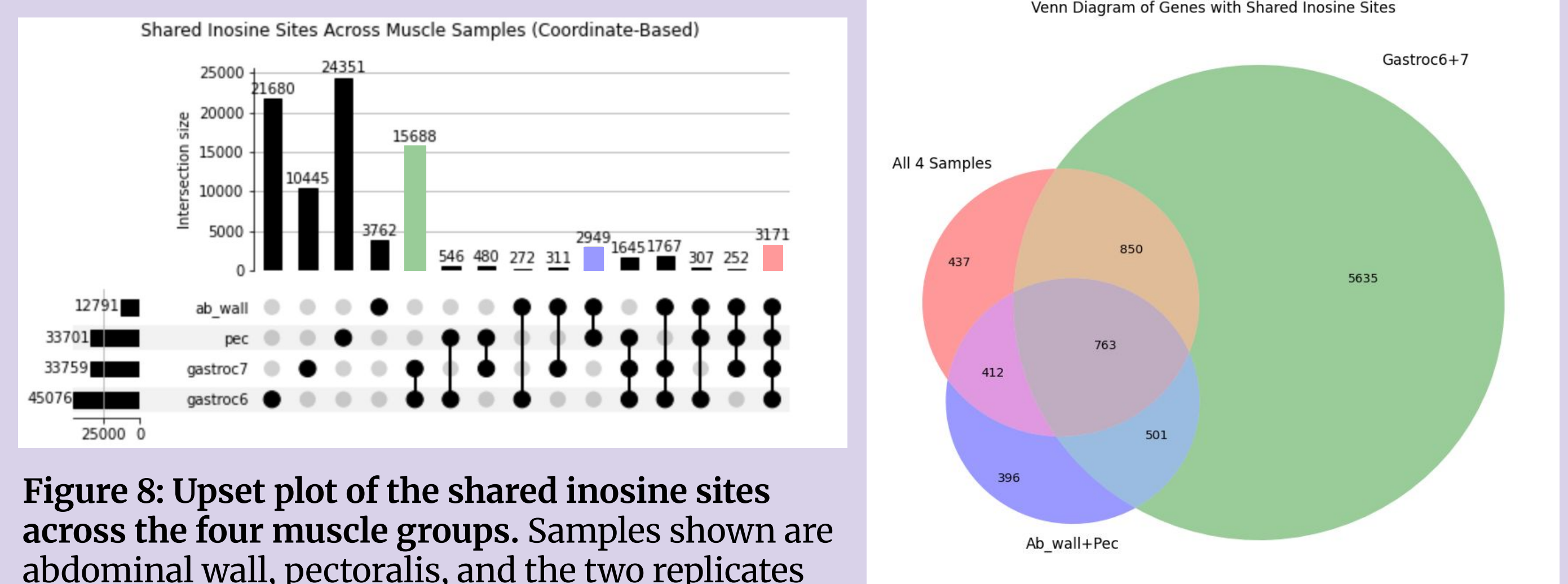


Figure 8: Upset plot of the shared inosine sites across the four muscle groups. Samples shown are abdominal wall, pectoralis, and the two replicates of gastrocnemius muscles.

Figure 9: Venn Diagram: Overlap of Inosine-Modified Genes. Each section without overlap indicates the inosine modifications specific to each group (all four, trunk muscle, and limb muscle). The overlaps indicate that the groups have at least 2 shared inosine modification sites (3 for grey).

Integrative Genomics Viewer of CKM Gene And GO Analysis

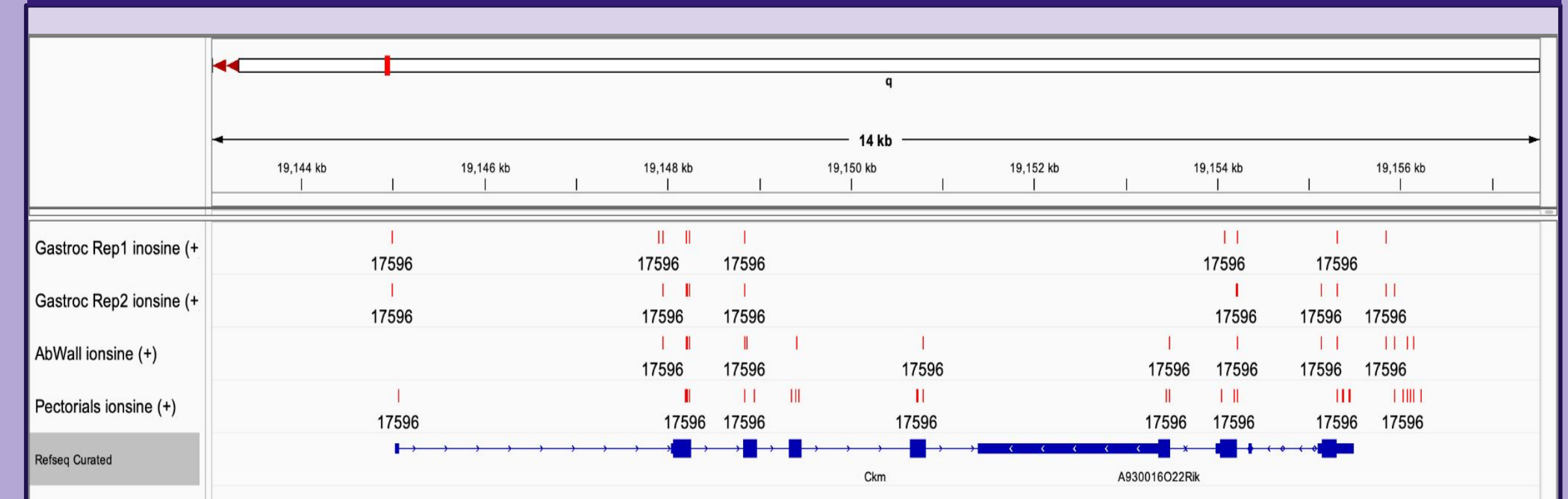


Figure 10: Integrative Genomics Viewer of Creatine kinase, M type (CKM). CKM, one of 763 overlapping genes shared between the muscle groups with inosine modifications at a same position. CKM is an important biomarker in muscular dystrophies. CKM is also the most variable gene expressed between limb and trunk muscles as seen in Figure 5.

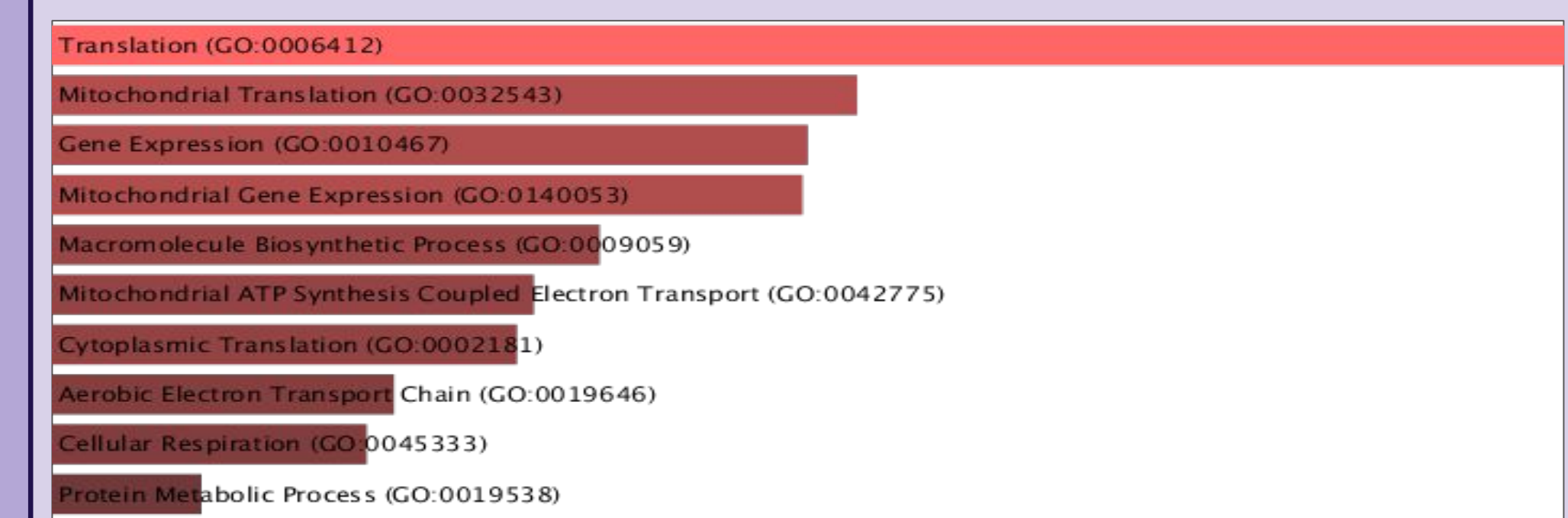


Figure 11: GO Analysis of all four muscle groups that have unique inosine modifications. The biological processes to the genes that have inosine sites specific to being present in all four muscle groups are significant—indicating that translation, more specifically, mitochondrial translation is unique to that group.

Conclusion & Future Directions

- Direct RNA sequencing of trunk and limb muscles in C57BL/6 mice enables precise comparison of gene expression and identification of inosine RNA modification sites.
- Both muscle groups share overlapping positions and counts of inosine modifications in genes linked to muscular dystrophy such as CKM gene.
- Despite shared modification sites, gene expression levels differ significantly between trunk and limb muscles.
- Trunk muscles display elevated expression of genes tied to cold-induced thermogenesis, while limb muscles show downregulation of calcium signaling and muscle contraction genes.
- Distinct inosine modification patterns indicate a role for post-transcriptional regulation in driving functional differences between trunk and limb muscle groups.
- RNA Fluorescence In Situ Hybridization (RNA FISH) will allow precise visualization of the spatial and cellular localization of transcripts with inosine modifications in muscle tissues.
- Compare muscle transcriptomic differences across similar development, aging, exercise, dystrophy mice models

References

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- Skeletal Muscle (<https://www.ncbi.nlm.nih.gov/books/NBK537236/>)
- RNA Modifications (<https://www.sciencedirect.com/science/article/pii/S0092867417306384>)
- Inosine Modifications (<https://pmc.ncbi.nlm.nih.gov/articles/PMC11019749/>)
- Gene Expression (<https://www.genome.gov/genetics-glossary/Gene-Expression#:~:text=Gene%20expression%20is%20the%20process,molecules%20that%20serve%20other%20functions>)
- Figure 1: Inosine Modifications (<https://pmc.ncbi.nlm.nih.gov/articles/PMC11019749/>)
- RNA FISH (<https://pmc.ncbi.nlm.nih.gov/articles/PMC7085896/>)

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